GAS LAWS

Gas is a state of matter with relatively low density and viscosity, variable pressure and temperature, and the ability to diffuse readily and distribute uniformly throughout any container. The gas laws described below relate movements of molecules within a gas to its volume, pressure, and temperature, and state how each measure responds when the others change. Most of the laws are named after the person who discovered them.

LAW	DESCRIPTION	FORMULA	KEY
Avogadro's law	At a constant temperature and pressure, volume is proportional to the number of molecules. Equal volumes of different gases, in identical conditions of temperature and pressure, will contain equal numbers of molecules.	$V \propto n$	V = volume n = number of molecules $\propto$ = proportional to
Boyle's law	For a given mass of gas at a constant temperature, volume is inversely proportional to pressure. So, for example, if volume doubles, pressure halves.	$P\,V$ = constant	P = pressure V = volume
Charles's law	For a given mass of gas at a constant pressure, volume is directly proportional to absolute temperature (measured in kelvin).	$\frac{V}{T}$ constant	V = volume T = (absolute) temperature
Gay-Lussac's law	For a given mass of gas at a constant volume, pressure is directly proportional to absolute temperature (measured in kelvin).	$\frac{P}{T}$ constant	P = pressure T = (absolute) temperature
Ideal gas law	An ideal gas is a hypothetical gas in which particles may collide but do not have attractive forces between them. The ideal gas law is a good approximation of the behavior of many gases in various conditions.	$P\,V = n\,R\,T$	P = total pressure n = number of moles R = universal gas constant T = absolute temperature
Dalton's law of partial pressures	This law describes mixtures of two or more gases. It states that the total pressure of a gaseous mixture equals sum of the partial pressures of individual component gases. The law is used to work out gas mixtures breathed by scuba divers.	$P = \Sigma p$ or P total $= p_1 + p_2 + p_3 \ldots$	P = total pressure Σp = sum of individual partial pressures

PRESSURE AND DENSITY

Pressure is the force applied to an object divided by the area over which the force is applied, and it varies according to the density of the body exerting the force. Pressure and density can be described using a series of equations.

LAW	DESCRIPTION	FORMULA
Pressure	$\frac{\text{force}}{\text{area}}$	$P = \frac{F}{A}$
Density	$\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
Volume	$\frac{\text{mass}}{\text{density}}$	$V = \frac{m}{\rho}$
Mass	volume × density	$m = V\rho$

EINSTEIN'S THEORIES OF RELATIVITY

In 1905 and 1915, German physicist Albert Einstein published his ground-breaking theories that challenged the previously accepted theory of gravitation.

THEORY	PROPOSITION
Special theory of relativity	1) All physical laws are the same in all frames of reference in uniform motion with respect to one another. 2) The speed of light is a constant, regardless of the motions of the light source and the observer.
General theory of relativity	Spacetime is curved; strong gravity causes distortions of time and mass, and large objects (such as stars) warp spacetime around them.

MAXWELL'S EQUATIONS

This is a series of equations, or laws, formulated by Scottish physicist James Clerk Maxwell (see 1855) that provide a full description of how electromagnetic waves behave. The equations show how electromagnetic fields are produced and how the rates of change in the fields are related to their sources.

LAW	STATEMENT	APPLICATION
Gauss's law for electricity	The electric flux through any closed surface is proportional to the total charge contained within that surface.	Used to calculate electric fields around charged objects.
Gauss's law for magnetism	For a magnetic dipole (one of a pair of equal and oppositely magnetized poles) with any closed surface, the magnetic flux drawn inward toward the south pole will equal the flux directed outward from the north pole; the net flux will always be zero.	Describes sources of magnetic fields and shows that they will always be closed loops.
Faraday's law of induction	The induced electromotive force (EMF) around any closed loop equals the negative of the rate of change of the magnetic flux through the area enclosed by the loop.	Describes how a changing magnetic field can generate an electric field; this is the operating principle for electric generators, inductors, and transformers.
Ampère's law with Maxwell's correction	In static electric field, the line integral of the magnetic field around any closed loop is proportional to the electric current flowing through it.	Used to calculate magnetic fields. Shows that magnetic current can be generated by electric current and by changing electric fields.